

Q1: CSP – Constraint Graph + Backtracking with Heuristics

Consider variables $\{A, B, C, D, E, F\}$ with domains:

- $A, B, C \in \{1, 2, 3, 4\}$
- $D, E, F \in \{2, 3, 4, 5\}$

Constraints:

- $A \neq B, A < C$
- $B + D = 6$
- $C \neq D$
- $D < E < F$
- $E \neq A$
- $F \neq B$

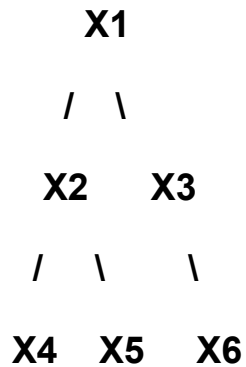
Tasks:

- Draw the constraint graph clearly (nodes and edges).
- Apply Backtracking Search using MRV + Degree Heuristic + Forward Checking.
- Show the search tree (partial assignments at each step).
- Give one consistent solution.

Q2: CSP – Tree-Structured Problem

Given a tree-structured CSP:

Structure:



Domains: All variables $\in \{1, 2, 3\}$

Constraints:

- $X1 \neq X2, X1 \neq X3$
- $X2 < X4$
- $X2 \neq X5$
- $X3 < X6$
- $X4 \neq X5$

Tasks:

- Convert this into a directed tree (choose root and order).
- Apply the Tree-CSP algorithm step-by-step.
- Show domain reductions at each stage.
- Provide final assignments.

Q3: Cryptarithmic

Solve:

CROSS + ROADS = DANGER

Tasks:

- Define variables and domains.

- **Write column-wise constraints including carries.**
- **Solve systematically (not by guessing).**
- **Show how carries propagate across columns.**